

Geometric Sequence / Series

$$ar^0 + ar + ar^2 + \cdots + ar^{n-1}$$
$$= \sum_{k=1}^n ar^{k-1} \quad \text{or} \quad \sum_{k=0}^{n-1} ar^k = \frac{a(1-r^n)}{1-r} \quad (r \neq 1).$$
$$\sum_{k=m}^n x^k = \frac{x^m - x^{n+1}}{1-x}, \quad (x \neq 1, \, m \leq n).$$

Simple Interest

$$a(t) = a(0) \, (1 + rt), \quad t \geq 0.$$

Bonds

Pays a specific amount (*face value / par value*) at maturity.
Most bonds pay periodic coupon payments.

Compound Interest

$$a(t) = a(0) \, (1 + r)^t, \quad t \geq 0.$$

Frequency of Compounding

If interest is compounded *p* times per year, denote the nominal rate by *r*^(*p*):

$$a(t) = a(0) \left(1 + \frac{r^{(p)}}{p}\right)^{tp},$$

where *t* is the number of years and *p* is the frequency of compounding.

Example. Let *a*(0) = 1000, nominal rate *r*⁽¹²⁾ = 0.06 (compounded monthly, *p* = 12), for *t* = 2 years.

$$a(2) = 1000 \left(1 + \frac{0.06}{12}\right)^{12 \cdot 2} = 1000(1.005)^{24} \approx 1127.16.$$

Effective Annual Interest Rate

$$1 + r_e = \left(1 + \frac{r^{(p)}}{p}\right)^p,$$

where *r_e* is the effective annual rate.

Two nominal rates are equivalent iff they have the same *r_e*.

Continuous Compounding

$$a(t) = \lim_{p \rightarrow \infty} a(0) \left(1 + \frac{r^{(\infty)}}{p}\right)^{tp} = a(0) e^{r^{(\infty)}t}.$$

Present Value & Future Value

$$FV = PV(1 + r)^n,$$

- FV* (**Future Value**): total amount at the end of the investment period (principal + accumulated interest).
- PV* (**Present Value**): initial amount you start with (principal investment).
- r* (**Interest Rate**): growth rate per period (decimal form, e.g. 5% = 0.05).
- n* (**Number of Periods**): total number of compounding periods (e.g. years, months, quarters).

Cash Flows

Money has *time value*: 1 baht today is worth more than 1 baht in the future because it can earn interest. Assume an effective annual interest rate *i* > 0 with discount factor

$$v = (1 + i)^{-1}.$$

Cash flows

Cash flow is a sequence of payments at times *t*₁, . . . , *t_n*:

$$(t_1, C_1), (t_2, C_2), \dots, (t_n, C_n),$$

where each *C_k* may be a *benefit* (*C_k* > 0) or *cost* (*C_k* < 0).

Present value (PV)

Present value at time 0:

$$PV = V(0) = \sum_{k=1}^n C_k v^{t_k}.$$

This captures *discounting*: future money is worth less today.

Value at a comparison time *s*

A payment *C* at time *t* has value at time *s*:

$$\text{Value at } s = C(1 + i)^{s-t}.$$

If *s* < *t* it is discounting; if *s* > *t* it is accumulation.

Equations of value & principle of equivalence

Choose a **comparison date** *s* and convert all payments to time *s*. For cash flows *A* and *B*, the **equation of value** is

$$V_A(s) = V_B(s).$$

Principle of equivalence: if two cash flows have the same value at one time under the same *i*, they have the same value at any other time under that same *i*.

Effective interest rate & Rule of 72

An unknown effective annual rate *i* can be solved by setting values equal (via an equation of value).

Rule of 72 (doubling time approximation):

$$n \approx \frac{0.72}{i},$$

where *i* is in decimal form (e.g. 0.12 for 12%). It is close for rates around 4%–12%.

Net present value (NPV)

Combine inflows/outflows into net payments *P_k* at times *t_k*. Then

$$NPV(i) = \sum_{k=1}^n P_k v^{t_k}.$$

Interpretation:

$$NPV(i) > 0 \Rightarrow \text{net gain (profitable),}$$

$$NPV(i) = 0 \Rightarrow \text{break-even,} \quad NPV(i) < 0 \Rightarrow \text{net loss.}$$

Yield / Internal rate of return (IRR)

IRR (yield) is the rate *i* such that

$$NPV(i)=0.$$

If borrowing rate < *i*, the project produces profit; if borrowing rate > *i*, it produces loss. If a cash flow has multiple sign changes, there may be multiple IRRs; in that case, NPV is preferred.

Inflation and tax adjustments

Inflation rate *j*: a basket costing 1 today costs (1 + *j*)^{*t*} after *t* years.

Real value (purchasing power) at time 0 of nominal amount *C* paid at time *t*:

$$C_{\text{real}}(0) = \frac{C}{(1 + j)^t}.$$

Real interest rate *r* (Fisher relation):

$$1 + i = (1 + r)(1 + j), \quad r = \frac{1 + i}{1 + j} - 1.$$

Real present value of payments (*t_k*, *C_k*):

$$V_{\text{real}}(0) = \sum_{k=1}^n C_k (1 + r)^{-t_k}.$$

Tax on interest: if tax rate on interest is *ω* ∈ [0, 1], then

$$i_{\text{after}} = i(1 - \omega).$$

Combining inflation and tax, the **real after-tax rate** is

$$r_{\text{after}} = \frac{1 + i_{\text{after}}}{1 + j} - 1.$$

Annuities (Certain)

Equal payments at regular intervals (fixed number of payments).
Values computed via geometric series.

Annuity-Immediate (payments at END of period)

Definition: Payments of 1 unit at times 1, 2, . . . , *n*. Let *i* = effective interest rate per period, $v = \frac{1}{1 + i}$.

Accumulated value at time *n*

$$s_n = 1 + (1 + i) + (1 + i)^2 + \cdots + (1 + i)^{n-1} = \frac{(1 + i)^n - 1}{i}.$$

Present value at time 0

$$a_n = v + v^2 + \cdots + v^n = \frac{1 - v^n}{i}.$$

Connection (same cashflows, different valuation time)

$$a_n = v^n s_n.$$

Annuity-Due (payments at BEGINNING of period)

Definition: Payments of 1 unit at times 0, 1, . . . , *n* − 1.

Key relationship (due vs immediate)

$$\ddot{a}_n = (1 + i)a_n.$$

Discount rate form

Define discount rate

$$d = \frac{i}{1 + i}, \quad 1 - v = d.$$

Then

$$\ddot{a}_n = \frac{1 - v^n}{d}, \quad \ddot{s}_n = \frac{(1 + i)^n - 1}{d}.$$

Useful shift identities

$$\ddot{a}_{n+1} = a_n + 1, \quad s_{n+1} = \ddot{s}_n + 1.$$

Common setups (quick use)

- Level-payment loan:** If loan amount is *L* repaid by *n* equal end-of-period payments *R*:

$$L = Ra_n \quad \Rightarrow \quad R = \frac{L}{a_n}.$$

- Savings plan:** If you deposit *R* each period (end), accumulated value at time *n*:

$$FV = Rs_n.$$

- Premiums/fees paid in advance (due):** Present value:

$$PV = R\ddot{a}_n.$$

Brownian Motion

Random Walk → Brownian Motion

For a symmetric random walk,

$$S_n = \sum_{i=1}^n X_i, \quad P(X_i = \pm 1) = \frac{1}{2},$$

so

$$E[S_n] = 0, \quad \text{Var}(S_n) = n.$$

Over horizon [0, *T*],

$$\Delta t = \frac{T}{n}, \quad \Delta x = \sqrt{\Delta t} = \sqrt{\frac{T}{n}},$$

which keeps total variance finite:

$$\text{Var}\left(\sum_{i=1}^n \sqrt{\frac{T}{n}} X_i\right) = T.$$

Continuous limit:

$$W^{(n)}(t) = \frac{1}{\sqrt{n}} S_{\lfloor nt \rfloor} \longrightarrow W(t).$$

Wiener Process

A process {*W*(*t*) : *t* ≥ 0} satisfies

$$W(0) = 0, \quad W(t) \sim N(0, t),$$

has continuous paths and independent increments. For 0 ≤ *s* < *t*,

$$W(t) - W(s) \sim N(0, t - s),$$

hence

$$E[W(t) - W(s)] = 0, \quad \text{Var}(W(t) - W(s)) = t - s.$$

- Continuous but nowhere differentiable.
- Spatial homogeneity: *W_x*(*t*) = *x* + *W*₀(*t*).
- Markov:** future depends on the present state, not the full past.
- Independent increments:** disjoint-time increments are independent.
- Martingale:**

$$E[W(t) \mid \mathcal{F}_s] = W(s), \quad s \leq t.$$

- Quadratic variation:

$$[W, W](t) = \lim_{\|\Pi\| \rightarrow 0} \sum_k (W(t_k) - W(t_{k-1}))^2 = t.$$

Brownian Motion with Drift

$$X(t) = X(0) + \mu t + \sigma W(t).$$

Therefore

$$X(t) \sim N(X(0) + \mu t, \sigma^2 t),$$

and for *t* > 0,

$$X(t + t_0) - X(t_0) \sim N(\mu t, \sigma^2 t).$$

Geometric Brownian Motion

$$S(t) = S(0)e^{\mu t + \sigma W(t)}.$$

Thus

$$\ln \frac{S(t)}{S(0)} \sim N(\mu t, \sigma^2 t),$$

so *S*(*t*) is **lognormal** and always positive.

Stochastic Calculus

Differential Wiener Rules

For a small interval dt ,

$$dW \sim N(0, dt), \quad E[dW] = 0.$$

Itô multiplication rules:

$$\boxed{(dW)^2 = dt}, \quad dt \, dW = 0, \quad (dt)^2 = 0.$$

Itô Integral

$$\int_0^t f(\tau) \, dW(\tau) = \lim_{n \rightarrow \infty} \sum_{k=1}^n f(t_{k-1}) (W(t_k) - W(t_{k-1})).$$

Left endpoints are used.

Linearity:

$$\int_0^t (f \pm g) \, dW = \int_0^t f \, dW \pm \int_0^t g \, dW,$$

$$\int_0^t c f \, dW = c \int_0^t f \, dW.$$

If f is deterministic and square-integrable,

$$E\left[\int_0^t f(\tau) dW(\tau)\right] = 0,$$

$$\text{Var}\left[\int_0^t f(\tau) dW(\tau)\right] = \int_0^t f(\tau)^2 \, d\tau.$$

Stochastic Differential Equation

$$dX(t) = \mu \, dt + \sigma \, dW(t)$$

has solution

$$X(t) = X(0) + \mu t + \sigma W(t),$$

so

$$E[X(t)] = X(0) + \mu t, \quad \text{Var}(X(t)) = \sigma^2 t.$$

Itô Process

General form:

$$dX = a(X, t) \, dt + b(X, t) \, dW.$$

$$\underbrace{a(X, t)}_{\text{drift}} \quad \underbrace{b(X, t)}_{\text{diffusion / volatility}}.$$

Itô's Lemma

If

$$dX = a \, dt + b \, dW, \quad Y = F(X, t),$$

then

$$\boxed{dY = \left(F_t + aF_x + \frac{1}{2}b^2F_{xx} \right) dt + bF_x \, dW}.$$

Useful Exponential Conversion

If

$$X = \ln P, \quad dX = \mu \, dt + \sigma \, dW,$$

then with $P = e^X$,

$$\boxed{\frac{dP}{P} = \left(\mu + \frac{1}{2}\sigma^2 \right) dt + \sigma \, dW}.$$

Conversely, if

$$\frac{dP}{P} = \alpha \, dt + \sigma \, dW,$$

then

$$d(\ln P) = \left(\alpha - \frac{1}{2}\sigma^2 \right) dt + \sigma \, dW.$$

Options

Core Terminology

- **Option**: right, not obligation, to buy/sell at strike K by expiry T .
- **Call**: right to **buy**. **Put**: right to **sell**.
- **European**: exercise only at T .
- **American**: exercise at or before T .
- **Long**: buy option. **Short**: sell/write option.

Payoffs at Expiry

Let $S_T = S(T)$.

Long Call = $C_K(S, T) = \max(S_T - K, 0)$,

Short Call = $-\max(S_T - K, 0)$,

Long Put = $P_K(S, T) = \max(K - S_T, 0)$,

Short Put = $-\max(K - S_T, 0)$.

Payoff vs Profit

If call premium is c and put premium is p :

$\Pi_{\text{long call}} = \max(S_T - K, 0) - c,$

$\Pi_{\text{short call}} = c - \max(S_T - K, 0),$

$\Pi_{\text{long put}} = \max(K - S_T, 0) - p,$

$\Pi_{\text{short put}} = p - \max(K - S_T, 0).$

Breakeven / Risk

Position	Breakeven	Max Gain	Max Loss
Long Call	$K + c$	unlimited	c
Short Call	$K + c$	c	unlimited
Long Put	$K - p$	$K - p$	p
Short Put	$K - p$	p	$K - p$

Moneyness

	ITM	ATM	OTM
Call	$S > K$	$S = K$	$S < K$
Put	$S < K$	$S = K$	$S > K$

Common Option Strategies

Protective Put:

Long Stock + Long Put,

payoff = $S_T + \max(K - S_T, 0) = \max(S_T, K)$.

Protects against a large fall in the asset price.

Straddle: same strike K , same expiry:

$+C_K + P_K,$

payoff = $\max(S_T - K, 0) + \max(K - S_T, 0) = |S_T - K|$.

Used when a **large move** is expected but direction is uncertain.

Strangle: long put and long call with **different strikes**.

Portfolio Payoff Construction

Option payoffs are additive. For example,

$\Pi(S_T) = -P_{50} + P_{100} + C_{100} - C_{150}.$

Each long call adds slope **+1** after its strike; each short call adds **−1**.

Puts create the analogous slope changes to the left of their strikes.

Put–Call Parity

For European options on a **non-dividend-paying** stock with the same K, T :

$$\boxed{S + P_K(S, t) = C_K(S, t) + Ke^{-r(T-t)}}.$$

Equivalent forms:

$C - P = S - Ke^{-r(T-t)},$

$P = C + Ke^{-r(T-t)} - S,$

$C = S + P - Ke^{-r(T-t)}.$

Continuous Discounting

If money grows at continuously compounded risk-free rate r ,

$M(t) = Ae^{rt}.$

Present value at time t of receiving K at T :

$PV = Ke^{-r(T-t)}.$